"Is Stephen Curry really a guard? New perspective on players typology using functional data analysis": Reviewer Responses

Steven Golovkine and Edward Gunning.

General/ Overall Responses

We thank the associate editor and the two reviewers for their thorough and detailed review of our manuscript.

Reviewer # 1

Comment # 1.1

The paper demonstrates the use of functional data analysis methods applied to shot chart densities in basketball for investigating common patterns among players. The shot chart is an important object of quantitative analysis in basketball, particularly when investigating spatial features in offensive strategies and performance. While the referees are in agreement that the paper targets a problem of interest, their reports contains several suggestions for revision. Below are some key issues with the paper, some of which overlap with comments from the two referees, that should be carefully and fully addressed in a revision, in addition to other points raised by the referees.

Reply:

Comment # 1.2

In the introduction, several methods for analyzing shot charts are summarized, including several based on intensity surfaces. The authors cite the requirement of a discretized grid for these methods as the reason for pursuing a density-based analysis, but this is not convincing. All functional data analyses must, in the end, be performed on a discrete grid, and it is not obvious that the estimated (log) intensity surfaces could not also be used as functional data objects. Another key difference between bivariate densities and intensity surfaces is that the latter contain information about shot volume, an aspect completely absent in the density representation. Please provide a more thorough justification for using densities as data objects as opposed to existing approaches.

Reply:

Comment # 1.3

While it is true that early work in functional data analysis did include applications of FPCA (or other standard FDA methods) directly to densities (Kneip & Utikal, 2001; Nerini & Ghattas, 2007; Ramsay & Silverman, 2005), these methods have been superseded in the last decade by non-linear dimension reduction methods due to the fact that density properties are not retained by linear decompositions (Bigot et al., 2017; Delicado, 2011; Petersen & Müller, 2016). Many if not all of these newer methods can only be applied to univariate densities; however, the authors should include a discussion about the potential pitfalls of FPCA when applied to

> *functional data that, by nature, satisfy nonlinear constraints (positivity and integration to 1). If possible, some attempt to mitigate these issues for the bivariate shot chart densities should be made.*

Reply:

COMMENT # 1.4

> *The choice of k-medioid clustering is also not discussed in detail. As suggested by the reviewers, please investigate the robustness or variability of your findings to more clustering methods, or provide a justification for this particular approach.*

Reply:

COMMENT # 1.5

> *A key drawback of the analysis is its lack of assessment of sampling variability. This is critical both for the reliability of the interpretation of the functional principal components (particularly those with smaller eigenvalues) as well as the clustering results. Without some inferential instrument (e.g., confidence bands), it is impossible to judge the importance of the findings.*

Reply:

COMMENT # 1.6

> *Although the analytical steps are clearly laid out, it is not obvious what the clustering should reflect relative to player positions. Based on the title and some of the conclusions stated by the authors, the underlying assumption seems to be that there is something unexpected or interesting about the lack of agreement between the clustering and designated player positions. However, there seems no obvious reason why these should coincide given different coaching styles and combination of player strengths. Please provide more concrete guidance about how the proposed analytical pipeline can be useful.*

Reply:

Comment # 1.7

> *Please carefully check the interpretation of each of the functional principal components. For instance, I do not believe it is the case that the first component reflects players who shoot more or less than average. This is because the functions being analyzed contain no information about overall shot volume, but only the distribution of shots. I also do not believe it is possible that the entire first component is positive, since it is known – at least in the univariate case (Kneip & Utikal, 2001) – that principal components for densities must integrate to zero.*

Reply:

Comment # 1.8

> *In addition to potentially comparing with additional clustering methods, further comparisons should be made between the proposed density-based approach and existing intensity-based approaches (functional or non-functional) in order to highlight advantages, if any, of the proposed pipeline.*

Reply:

Reviewer # 2

Comment # 2.1

The current research is characterizing NBA player shooting behavior using functional data analysis. The authors treat made and missed shot locations as bivariate functional data on the offensive half-court and perform a multivariate functional principal component analysis to extract dominant shooting patterns. Then, using k-medoids clustering on the resulting player-specific scores, they generate a data-driven typology of player shooting styles, questioning the validity of traditional NBA positional labels. Several conceptual and technical issues limit both the novelty and the inferential credibility of the results. I therefore recommend a revision before the paper can be considered for publication.

Reply:

Comment # 2.2

While the paper is intended for a quantitative audience, some statistical steps, especially the MFPCA, are described at a relatively informal level. There is no discussion of convergence or uncertainty in the functional principal components.

Reply:

Comment # 2.3

The authors do not evaluate the stability of their clustering. Furthermore, they do not provide external validation (e.g., prediction, test–train splits, comparison with existing clustering approaches). Please consider using metrics such as Adjusted Rand Index or Silhouette scores to assess cluster quality. Alternatively, compare results with simpler clustering methods (e.g., k-means on shooting frequency by location bins) to demonstrate added value.

Reply:

Comment # 2.4

While contrasting the clusters with NBA-defined positions is informative, the resulting disagreement is not deeply analyzed. Are there specific cases where agreement is worse?

Reply:

COMMENT # 2.5

The paper uses a 2D KDE with Gaussian kernel and Silverman's rule but does not specify whether the bandwidth is optimized globally, per player, or per component.

Reply:

COMMENT # 2.6

The model jointly treats made and missed shots, but no attention is given to efficiency metrics (i.e., shooting percentage), which are arguably central to evaluating shot quality.

Reply:

Reviewer # 3

Comment # 3.1

> *The article is interesting and has an interesting premise but needs additional work. I have included a number of grammatical edits below. Please make these corrections and ask a peer to review the article for other edits that I may have missed. I included non-grammatical comments below the grammatical fixes. P2, L23: "his/her" not its P2, L29: "to estimate" P3, L22-23: inconsistent quotations P4, L40: characterizes P4, L50: capitalize Euclidean P5, L36: rewrite sentence L38: "short corners?" (There are only two corners and they are both short.) 41: English 42: English 43: will "shoot" Break up the first paragraph in 4.1 into multiple smaller ones.*

Reply:

We thank the reviewer for their comments. We corrected the grammatical errors. Regarding the "short corners", the "short corners" refers to specific location on the court. We added a graphic that explain the different names of the court location.

Comment # 3.2

> *Can you explain how this is shown by the 3rd PC? "Interestingly, players who perform well near the basket also tend to attempt and miss shots from the corners"*

Reply:

Comment # 3.3

> *Last sentence (P5, L55-56) does not make sense. What is "the axis of the hoop" or "the axis of the basket" (you use both terms)?*

Reply:

It refers to shooting in front of the basket as oppose to shooting in the wings/corners.

Comment # 3.4

> *Figure 5: for visualization purposes, consider normalizing the scores so that the axes all line up. Also consider adding a vertical line at $x = 0$ so you can more easily determine if the score is positive or negative. If possible, caption each graph with the interpretation you are assigning to it so that a reader can look at the graph and understand what it means for the*

labeled players.

Reply:

COMMENT # 3.5

P6 L51: "designated" as outliers

Reply:

We thank the reviewer for spotting the typo.

COMMENT # 3.6

P9 L51: describe the clusters in numeric order.

Reply:

COMMENT # 3.7

P10: give a link / reference to Hawk-Eye; "player's"

Reply:

COMMENT # 3.8

At the very least, you should report the of guards, forwards, and centers in each cluster. (Many readers may not be able to distinguish who is a guard or who is a forward, etc, just by reading players' names.) It would also be nice if you used a quantitative metric to measure the agreement between the clusters you find and the NBA positions. If you try other functional clustering methods (as I suggest below), these comparisons would be even more interesting.

Reply:

COMMENT # 3.9

Did you consider other functional clustering approaches? It would be interesting to see if you get similar clusters (and similar levels of agreement) using a different approach.

COMMENT # 3.10

Why not cluster the surfaces themselves and not the MFPCA scores using functional clustering algorithms? That would be fun to see.

COMMENT # 3.11

What about just FPCA of all shot locations (as opposed to made and missed shots?) Do you gain anything from doing MFPCA in the way that you do vs just FPCA of shot locations?

Reply:

References

Bigot, J., Gouet, R., Klein, T., & López, A. (2017). Geodesic PCA in the Wasserstein space by convex PCA. *Annales de l'Institut Henri Poincaré, Probabilités et Statistiques*, *53*(1), 1–26.

Delicado, P. (2011). Dimensionality reduction when data are density functions. *Computational Statistics & Data Analysis*, *55*(1), 401–420.

Kneip, A., & Utikal, K. J. (2001). Inference for Density Families Using Functional Principal Component Analysis. *Journal of the American Statistical Association*, *96*(454), 519–532.

Nerini, D., & Ghattas, B. (2007). Classifying densities using functional regression trees: Applications in oceanology. *Computational Statistics & Data Analysis*, *51*(10), 4984–4993.

Petersen, A., & Müller, H.-G. (2016). Functional data analysis for density functions by transformation to a Hilbert space. *The Annals of Statistics*, *44*(1), 183–218.

Ramsay, J. O., & Silverman, B. W. (2005). *Functional Data Analysis*. Springer.